

CSE 202 Midterm 2 Part (a)

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I. QUESTION (QUEUE) (15 POINTS)

Write a function that deletes the element in the K 'th ($K \geq 0$) position of the queue. Write the function for array implementation.

```
void deleteKth(int K)
```

II. QUESTION (BINARY SEARCH TREE) (15 POINTS)

Write a recursive function that returns the number of nodes in a binary search tree whose data is divisible by 3.

III. QUESTION (BINARY SEARCH TREE) (15 POINTS)

Modify the `TreeNode` data structure such that each node has four children, namely, `left`, `leftOfMiddle`, `rightOfMiddle` and `right`. Then write the function search given that the search tree property is satisfied with this kind of tree.

IV. QUESTION (HASHING) (15 POINTS)

Write function that finds the number of empty slots in a hash table (For both array and linked list implementations).

```
int numberOfEmptySlots()
```